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Vellore Institute of Technology

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School of Computer Science and Engineering

BCSE305L – Embedded Systems

Module ~ IV Programming Tools

Logic Analyzer

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Logic Analyzer

Logic Analyzer

- A logic analyzer is an **electronic instrument** that captures and displays multiple signals from a digital system or a digital circuit.
- It is an **excellent tool for verifying and debugging digital designs**.
- A logic analyzer may convert the captured data into **timing diagrams, protocol decodes, state machine traces, and assembly language**.
- For debugging elusive, intermittent problems, some logic analyzers can detect glitches, as well as setup-and-hold time violations.
- During software/hardware integration, logic analyzers trace the execution of the embedded software and analyze the efficiency of the program's execution.
- Some logic analyzers correlate the source code with specific hardware activities in your design.

Why use a logic analyzer?

- Debugging **microprocessor-based designs** required more inputs than what conventional analog oscilloscopes could offer.
- A typical **logic analyzer** has anywhere from 8 to 136 channels and they are particularly useful for looking at time relationships or data on a bus – for example, a microprocessor address, data, or control bus.
- They can decode the information on microprocessor buses and display it in a meaningful form.

A logic analyzer is used when we need to:

- Debug and verify digital system operation
- Trace and correlate many digital signals simultaneously
- Detect and analyze timing violations and transients on buses
- Trace embedded software execution

Types of Logic Analyzers

There are three types of logic analyzers:

- Modular logic analyzers,
- Portable logic analyzers, and
- PC-based logic analyzers.

Modular Logic Analyzers:

- Modular logic analyzers are the standard form seen in labs that have a chassis and multiple modules.
- These are one of the more expensive and provide the highest level of functionality to the user.
- Modules can be added or removed depending on the user to increase the functionality.
- Eg: Modules can be added to increase the number of channels in the Logic Analyzer depending on the user.

Portable Logic Analyzer:

- Portable logic analyzers are more portable than modular logic analyzers and provide all the functions that are integrated into a single module with a screen.
- There may be instances when a smaller analyzer will be required due to restricted budgets or fields of service.
- These test instruments incorporate all elements of the analyzer into a single unit for ease of transportation.

PC-based Logic Analyzer:

- PC-based logic analyzers are compact and they directly interface to a computer via an ethernet or a USB cable.
- The captured information is displayed to the user via the PC's display.
- PC-based logic analyzers are the least expensive but are limited in terms of power compared to modular and portable logic analyzers.

- **Prodigy's Logic Analyzer** supports Real-time I3C (Improved Inter Integrated Circuit) Trigger & Protocol Analysis which enables engineers to debug timing, circuit-level, and system-level problems quickly.
- Advanced trigger, streaming, and unlimited data capture capabilities make it a must-have tool in your lab to debug I3C protocol.
- Perform simultaneous protocol analysis of #I3C, #I2C, #SPI, #UART, #SPMI, #CAN/CAN FD, and #RFFE in embedded designs.
- Features such as 16 channels, 1GS/s Timing Speed, 2ns glitch Analysis & 100MHz state data capture make it an ideal must-have debug tool to address digital design problems.
- Designers can now easily analyze setup and hold time issues, glitches, and synchronous data activities apart from analyzing protocol issues.